BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Strings: Concatenation and Slicing

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does assignment do to a variable?

2. What is NOT true?

A) true
B) 0.5
C) false
D) An empty string

3. What values does a Boolean variable represent?

4. Computers share files on one school site. Two sites in different towns also need a link. Correct both claims. Explain why. (a) A LAN spans several countries. (b) A WAN only works in one room.

Today's objectives

Manipulate strings using concatenation and substring operations. Use the technique in an independent scenario.

Key vocabulary - Read, cover, recall

Concatenation
The + operator joins two strings in order without automatically inserting a space.

Substring
substring(start,length) returns characters beginning at a zero-based index.

Length
The .length property gives the number of characters in a string.

Indexing
The first character in OCR ERL is at position zero.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Concatenation	The + operator joins two strings in order without automatically inserting a space. Apply this in a welcome message.
Substring	substring(start,length) returns characters beginning at a zero-based index. Apply this in a name badge.
Length	The .length property gives the number of characters in a string. Apply this in a club code.
Indexing	The first character in OCR ERL is at position zero. Apply this in a ticket suffix.

I Do - One worked example

A welcome message is being developed. Explain how concatenation and substring are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1].

3. Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1].

4. Two linked points for each concept.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A name badge uses concatenation. Explain how it helps. [2 marks]

2. Explain substring in a club code. [2 marks]

3. A ticket suffix uses length and indexing. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the substring to display the first three characters of "NETWORK".

```
name = "NETWORK"
print(name.substring(0, ...))
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State TWO relevant techniques from this topic. [2 marks]

Q3. Explain how substring and length help with a school password hint. [4 marks]

Q4. A programmer designs a café menu label. Explain why concatenation, length and indexing matter. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. The + operator joins two strings in order without automatically inserting a space. ☐ T ☐ F
2. The + operator joins two strings in order without automatically inserting a space ☐ T ☐ F
3. substring(start,length) returns characters beginning at a zero-based index ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 It stores a value in the named variable, replacing its previous value if one existed. ☐

Q2 C - false ☐

Q3 True or false. ☐

Q4 (a) A LAN covers a small area, such as a school site. (b) A WAN links networks over a large area, often using outside providers' links. Internet access alone does not decide the type. ☐

Code First - Model

```
name = "NETWORK"
print(name.substring(0, 3))
```

Output: NET

We Do - Check the model

1. Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1]. ☐

2. Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1]. ☐

3. Length [1]; The .length property gives the number of characters in a string. [1]. Indexing [1]; The first character in OCR ERL is at position zero. [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Concatenation [1]; Length [1]. ☐

Q3 Substring [1]; substring(start,length) returns characters beginning at a zero-based index. [1]. Length [1]; The .length property gives the number of characters in a string. [1]. ☐

Q4 Concatenation [1]; The + operator joins two strings in order without automatically inserting a space. [1]. Length [1]; The .length property gives the number of characters in a string. [1]. Indexing [1]; The first character in OCR ERL is at position zero. [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: